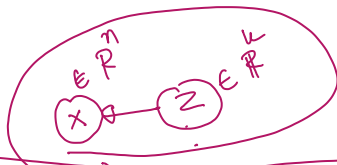


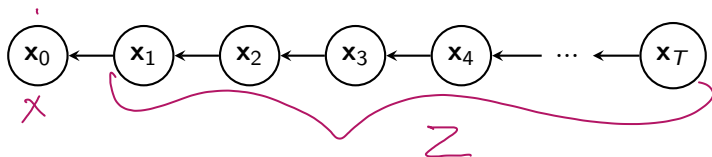
Structured Variational Approximation

$$\max_{\theta} \sum_{i=1}^N \log P_{\theta}(\mathbf{x}^i)$$



$$= \max_{\theta} \underbrace{\sum_{i=1}^N \max_{q(\mathbf{z}|\mathbf{x}^i)} \int_{\mathbf{z}} (q(\mathbf{z}|\mathbf{x}^i) \log P_{\theta}(\mathbf{x}^i|\mathbf{z}))}_{\text{Expected Likelihood}} - \underbrace{\int_{\mathbf{z}} q(\mathbf{z}|\mathbf{x}^i) \log \frac{q(\mathbf{z}|\mathbf{x}^i)}{P_{\theta}(\mathbf{z})} d\mathbf{z}}_{\text{KL-distance}(q(\mathbf{z}|\mathbf{x}^i), P_{\theta}(\mathbf{z}))}$$

Assume that $P(\mathbf{x}|\mathbf{z})$ is further factorized as a chain Bayesian network as follows where $\mathbf{x} \mapsto \mathbf{x}_0$, $\mathbf{z} \mapsto \mathbf{x}_1, \dots, \mathbf{x}_T$



Simplifying objective when $P_\theta(\mathbf{x}, \mathbf{z})$ is structured.

$$P_\theta(\mathbf{x}, \mathbf{z}) = P_\theta(x_0, x_1, x_2 \dots x_T) \\ = P(x_T) \prod_{t=0}^{T-1} P_\theta(x_t | x_{t+1})$$

$$\int_{\mathbf{z}, \mathbf{y}} p(\mathbf{x}, \mathbf{y}) g(\mathbf{y}) \\ = \int_{\mathbf{y}} (p(\mathbf{y}) g(\mathbf{y})) p(\mathbf{x} | \mathbf{y}) \\ = \int_{\mathbf{y}} p(\mathbf{y}) g(\mathbf{y}) d\mathbf{y}$$

$$P_\theta(\mathbf{x} | \mathbf{z}) = P_\theta(x_0 | x_1) \\ P_\theta(\mathbf{z}) = P(x_T) \prod_{t=1}^{T-1} P_\theta(x_t | x_{t+1})$$

Simplifying

$$\int_{\mathbf{z}} q(\mathbf{z} | \mathbf{x}) \log P(\mathbf{x} | \mathbf{z}) = \int_{\mathbf{x}_1 \dots \mathbf{x}_T} \underbrace{q(\mathbf{x}_1 \dots \mathbf{x}_T | x_0)}_{\log} \underbrace{P(x_0 | \mathbf{x}_1 \dots \mathbf{x}_T)}_{\log} d\mathbf{x}_1 \dots \mathbf{x}_T$$

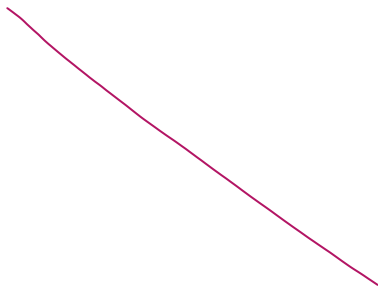
$$= \int_{\mathbf{x}_1 \dots \mathbf{x}_T} q(\mathbf{x}_1 \dots \mathbf{x}_T | x_0) \log P(x_0 | \mathbf{x}_1) d\mathbf{x}_1 \dots \mathbf{x}_T \\ = \int_{\mathbf{x}_1} q(\mathbf{x}_1 | x_0) P(x_0 | \mathbf{x}_1) d\mathbf{x}_1$$

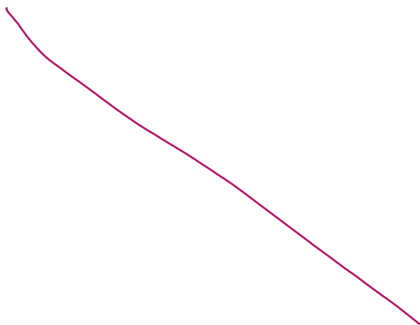
$$\min_{\mathbf{z}} \int q(\mathbf{z}|\mathbf{x}) \log \frac{q(\mathbf{z}|\mathbf{x})}{\underline{\underline{p(\mathbf{z})}}} = \int_{\mathbf{x}_1 \dots \mathbf{x}_T} q(\mathbf{x}_1 \dots \mathbf{x}_T | \mathbf{x}_0) \log \frac{q(\mathbf{x}_1 \dots \mathbf{x}_T | \mathbf{x}_0)}{p(\mathbf{x}_T) \prod_{t=1}^T p(\mathbf{x}_t | \mathbf{x}_{t+1})}$$

$$\underline{\underline{q(\mathbf{x}_1 \dots \mathbf{x}_T | \mathbf{x}_0)}} = \prod_{t=1}^{T-1} q(\mathbf{x}_t | \mathbf{x}_{t+1}, \mathbf{x}_0) \underline{\underline{q(\mathbf{x}_T | \mathbf{x}_0)}}$$

$$\rightarrow \sum_{t=1}^{T-1} \int_{\mathbf{x}_1 \dots \mathbf{x}_T} q(\mathbf{x}_1 \dots \mathbf{x}_T | \mathbf{x}_0) \log \frac{q(\mathbf{x}_t | \mathbf{x}_{t+1}, \mathbf{x}_0)}{p(\mathbf{x}_t | \mathbf{x}_{t+1})} + \underbrace{\int_{\mathbf{x}_1 \dots \mathbf{x}_T} q(\mathbf{x}_1 \dots \mathbf{x}_T | \mathbf{x}_0) \log \frac{q(\mathbf{x}_T | \mathbf{x}_0)}{p(\mathbf{x}_T)}}_{\text{wavy line}}$$

$$= \sum_{t=1}^T \underline{\underline{E_q [KL(q(\mathbf{x}_t | \mathbf{x}_{t+1}, \mathbf{x}_0); p(\mathbf{x}_t | \mathbf{x}_{t+1}))]} + E_q [KL(q(\mathbf{x}_T | \mathbf{x}_0); p(\mathbf{x}_T))]}_{\text{wavy line}}$$





Simplifying the above I *[Same stuff in cleaner notation]*

Note, $P(\mathbf{x}|\mathbf{z}) = P(\mathbf{x}_0|\mathbf{x}_1, \dots, \mathbf{x}_T) = P(\mathbf{x}_0|\mathbf{x}_1)$ [conditional independence in the above Bayesian network]

Thus, the first term becomes:

$$\int_{\mathbf{z}} q(\mathbf{z}|\mathbf{x}^i) \log P_{\theta}(\mathbf{x}^i|\mathbf{z}) = \int_{\mathbf{x}_1} q(\mathbf{x}_1|\mathbf{x}_0^i) \log P_{\theta}(\mathbf{x}_0^i|\mathbf{x}_1)$$

Note, also that

$$P_{\theta}(\mathbf{z}) = P_{\theta}(\mathbf{x}_1, \dots, \mathbf{x}_T) = P(\mathbf{x}_T) \prod_{t=1}^{T-1} P_{\theta}(\mathbf{x}_t|\mathbf{x}_{t+1})$$

Simplifying the above II

The second term, KL-distance($q(\mathbf{z}|\mathbf{x}^i), P_\theta(\mathbf{z})$) is minimized when the two distributions are equal. Since, $P_\theta(\mathbf{z})$ factorizes as above, we can show that an optimal $q(\mathbf{z}|\mathbf{x})$ will also factorize in the same way. Thus.

$$q(\mathbf{z}|\mathbf{x}) = q(\mathbf{x}_1, \dots, \mathbf{x}_T|\mathbf{x}_0) = q(\mathbf{x}_T|\mathbf{x}_0) \prod_{t=1}^{T-1} q(\mathbf{x}_t|\mathbf{x}_{t+1}, \mathbf{x}_0)$$

Overall KL divergence term becomes

$$\begin{aligned}& \int_{\mathbf{z}} q(\mathbf{z}|\mathbf{x}) \log \frac{q(\mathbf{z}|\mathbf{x})}{P_{\theta}(\mathbf{z})} d\mathbf{z} \\&= \int_{\mathbf{x}_1, \dots, \mathbf{x}_T} q(\mathbf{x}_1, \dots, \mathbf{x}_T|\mathbf{x}_0) \left(\log \frac{q(\mathbf{x}_T|\mathbf{x}_0)}{P(\mathbf{x}_T)} + \sum_{t=1}^{T-1} \log \frac{q(\mathbf{x}_t|\mathbf{x}_{t+1}, \mathbf{x}_0)}{P_{\theta}(\mathbf{x}_t|\mathbf{x}_{t+1})} \right) d\mathbf{x} \\&= \int_{\mathbf{x}_T} q(\mathbf{x}_T|\mathbf{x}_0) \log \frac{q(\mathbf{x}_T|\mathbf{x}_0)}{P(\mathbf{x}_T)} d\mathbf{x}_T \\&\quad + \sum_{t=1}^{T-1} \int_{\mathbf{x}_t, \mathbf{x}_{t+1}} q(\mathbf{x}_t, \mathbf{x}_{t+1}|\mathbf{x}_0) \log \frac{q(\mathbf{x}_t|\mathbf{x}_{t+1}, \mathbf{x}_0)}{P_{\theta}(\mathbf{x}_t|\mathbf{x}_{t+1})} d\mathbf{x}_t \\&= KL(q(\mathbf{x}_T|\mathbf{x}_0); P(\mathbf{x}_T)) + \sum_{t=1}^{T-1} \mathbb{E}_q[KL(q(\mathbf{x}_t|\mathbf{x}_{t+1}, \mathbf{x}_0); P_{\theta}(\mathbf{x}_t|\mathbf{x}_{t+1}))]\end{aligned}$$

Overall Structured Variational Objective I

$$\begin{aligned} \max_{\theta, q} & \sum_{i=1}^N \mathbb{E}_{\mathbf{x}_0 \sim D} \mathbb{E}_q [\log P_{\theta}(\mathbf{x}_0 | \mathbf{x}_1)] \\ & + \text{KL}(q(\mathbf{x}_T | \mathbf{x}_0); P(\mathbf{x}_T)) \\ & + \sum_{t=1}^{T-1} \text{KL}(q(\mathbf{x}_t | \mathbf{x}_{t+1}, \mathbf{x}_0); P_{\theta}(\mathbf{x}_t | \mathbf{x}_{t+1})) \end{aligned}$$

Diffusion models define a $q(\mathbf{x}_t | \mathbf{x}_{t+1}, \mathbf{x}_0)$ manually as follows:

- 1 Define $q(\mathbf{x}_0, \dots, \mathbf{x}_T) = q(\mathbf{x}_0) \prod_{t=1}^T q(\mathbf{x}_t | \mathbf{x}_{t-1})$. This is called a forward process.
- 2 Choose a simple fixed form for $q(\mathbf{x}_{t+1} | \mathbf{x}_t)$ e.g Gaussian or Multinomial such that the following three conditions are met:

Overall Structured Variational Objective II

- ① It is easy to compute

$$\underline{q(\mathbf{x}_t | \mathbf{x}_{t+1}, \mathbf{x}_0)} = \frac{q(\mathbf{x}_{t+1} | \mathbf{x}_t) q(\mathbf{x}_t | \mathbf{x}_0)}{\int_{\mathbf{x}} \underline{q(\mathbf{x}_{t+1} | \mathbf{x}) q(\mathbf{x} | \mathbf{x}_0)}}$$

- ② It is easy to compute $\underline{q(\mathbf{x}_t | \mathbf{x}_0)}$
- ③ $\underline{q(\mathbf{x}_T | \mathbf{x}_0)} = \underline{P(\mathbf{x}_T)}$